PART 4 - CAPSTONE PROJECT #6

Presentation Script & Audit Log

Tasks 67–72: Slide Narration Script, Automation Pipeline, Prompt Library & Verification Audit.

1. Presentation Script & Narration (2-3 Mins)

Slide 1: Capstone Introduction

> ***Script (0:00 - 0:30):*** *"Welcome to our ECE Capstone Project on AI Communication Systems. We explore how AI models accelerate learning modulation techniques and simulate waveforms."*

Slide 2: Analog Modulation (AM vs FM)

> ***Script (0:30 - 1:15):*** *"AM varies carrier amplitude while FM varies frequency. FM maintains constant peak power, making it superior against channel noise."*

Slide 3: Digital Keying (ASK vs FSK)

> ***Script (1:15 - 2:00):*** *"ASK pulses carrier ON/OFF for digital bits, whereas FSK shifts between two distinct frequencies."*

2. Quiz Automation Pipeline (Zapier/Make)

Workflow: Google Sheets (New Topic) → Zapier Webhook → OpenAI API Prompt Call → JSON Quiz Parser → Google Form Creation.

3. Prompt Repository & Verification Audit

Task	Prompt Executed	Audit Result / Verification
Step 63	"Explain AM, FM, ASK, FSK with math equations"	✓ **Verified against Haykin textbook**
Step 65	"Write Python code for AM, FM, ASK waveforms"	✓ **Code executed cleanly in Colab**
Step 66	"Generate 15 MCQ questions with answer key"	✓ **Corrected minor subscript index**

4. Final Submission Checklist (100 Marks Weightage)

- **PDF 1:** Verified Theory & Equations (Steps 61-64)
- **PDF 2:** Python Simulation Code & Waveforms (Step 65)
- **PDF 3:** Glossary, 15 MCQs & Answer Key (Step 66)
- **PDF 4:** Presentation Script, Automation & Audit Log (Steps 67-72)